

## Literature Benchmark Brief: Shazeer\_2017\_ICLR

**Title:** Outrageously large neural networks: The sparsely-gated Mixture-of-Experts layer

**Authors:** N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, J. Dean (2017)

**Venue:** ICLR | **DOI:** 10.48550/arXiv.1701.06538

**Cluster:** Cluster E: Missing Data & Modular Risk Layers

### Core Theoretical Mechanism & Model Formulation

Top-K gating, noisy softmax routing, auxiliary load-balancing loss

### Empirical Scale & Experimental Dataset

Large-scale language modeling and translation benchmarks

### Relevance to IEEE TSTE & Sustainable Energy Systems

Establishes dynamic MoE architecture and gating mechanics

### Identified Gaps & Operational Vulnerabilities

Unconstrained loss-driven routing lacks physical semantic interpretability

### Role in Manuscript Defense & Reviewer #2 Proof

Used as ablation comparator; manuscript proves dynamic MoE confers no benefit over Dense